THE FROBENIUS SERIES METHOD

(I) Consider an ODE: $\left[y'' + \frac{p(x)}{x} y' + \frac{q(x)}{x^2} y = 0\right]$

If $x=0$ is a Regular singular point, then

$p(x) = p_0 + p_1 x + p_2 x^2 + p_3 x^3 + \cdots$

$q(x) = q_0 + q_1 x + q_2 x^2 + q_3 x^3 + \cdots$

are analytic at $x=0$.

(II) The ODE has a Frobenius series solution:

$$y = x^r \sum_{n=0}^{\infty} c_n x^n \quad \text{OR} \quad y = \sum_{n=0}^{\infty} c_n x^{n+r}$$

where $c_0 \neq 0$

$$\Rightarrow \quad y' = \sum_{n=0}^{\infty} (n+r) c_n x^{n+r-1}$$

$$\Rightarrow \quad y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) c_n x^{n+r-2}$$

(III) Substitute the series solution into the ODE & find the Indicial Equation:

$$y'' + \frac{p(x)}{x} y' + \frac{q(x)}{x^2} y = 0$$

$$\Rightarrow x^2 y'' + x p(x) y' + q(x) y = 0 \longrightarrow \left[\text{After Multiplying by } x^2\right]$$

$x^2y'':\ x^2\sum_{n=0}^{\infty} c_n(n+r)(n+r-1)x^{n+r-2}$

$\Rightarrow \sum_{n=0}^{\infty} c_n(n+r)(n+r-1)x^{n+r}$

$x\cdot p(x)y':\ x[p_0+p_1x+p_2x^2+p_3x^3+\dots]$

$\sum_{n=0}^{\infty} c_n(n+r)x^{n+r-1}$

$\Rightarrow [p_0x+p_1x^2+p_2x^3+\dots]\left[\sum_{n=0}^{\infty} c_n(n+r)x^{n+r-1}\right]$

$q(x)y:\ [q_0+q_1x+q_2x^2+q_3x^3+\dots]\left[\sum_{n=0}^{\infty} c_n x^{n+r}\right]$

$x^2y''\big|_{n=0,1,2,\dots} = [r(r-1)c_0x^r+(r+1)rc_1x^{r+1}+\dots]$

$+$

$x\,p(x)y'\big|_{n=0,1,2,\dots} = [p_0x+p_1x^2+\dots][rc_0x^{r-1}+(r+1)c_1x^r+\dots]$

$+$

$q(x)y\big|_{n=0,1,2,\dots} = [q_0+q_1x+\dots][c_0x^r+c_1x^{r+1}+\dots]$

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①

Add + Collect the lowest-degree terms, x^r-terms

$$[r(r-1)c_0x^r]+(p_0x)(rc_0x^{r-1})+(q_0)(c_0x^r)=0$$

$$(r)(r-1)c_0x^r + rp_0c_0x^r + q_0c_0x^r = 0$$

$$\Rightarrow c_0[r(r-1) + rp_0 + q_0]x^r = 0$$

$\because c_0 \neq 0$

$$r(r-1) + rp_0 + q_0 = 0$$

{where $p_0 = p(0)$ & $q_0 = q(0)$} → INDICIAL EQUATION.

→ If our frobenius series is a solution to the ODE,

$$\left[\sum_{n=0}^{\infty} c_n x^{n+r}\right]$$

then, "r" in the Frobenius series should be one of roots of the indicial equation (r_1, r_2) $[r(r-1) + rp_0 + q_0 = 0]$.

Note: ① r_1, r_2: can be distinct roots such that $r_1 \geq r_2$

then $y_1(x) = x^{r_1}\sum_{n=0}^{\infty} a_n x^n$ is one solution $(a_0 \neq 0)$

② r_1, r_2: can be distinct roots such that
$(r_1 - r_2) \neq 0$
$(r_1 - r_2)$ is not a positive integer

then $y_2(x) = x^{r_2}\sum_{n=0}^{\infty} b_n x^n$ is a second, linearly independent solution

$(b_0 \neq 0)$

EXAMPLE: FINDING THE FROBENIUS SERIES

Find the Frobenius series solution of the equation

$$2x^2(1+x)y'' + 3x(1+x)^3 y' - (1-x^2)y = 0$$

Solution:

STEP I Write the ODE in standard form:

$$y'' + \frac{3x(1+x)^3}{2x^2(1+x)}y' - \frac{(1-x^2)}{2x^2(1+x)}y = 0$$

[Form: $y'' + P(x)y' + Q(x)y = 0$]

$$\Rightarrow y'' + \frac{\left(\frac{3}{2}\right)\cancel{x}(1+x)^{3-2}}{\cancel{x}}y' + \frac{\left(-\frac{1}{2}\right)(1-x)(1+x)}{x^2}y = 0$$

$$\Rightarrow y'' + \frac{\left(\frac{3}{2}\right)(1+x)^2}{x}y' + \frac{\left(-\frac{1}{2}\right)(1+x)(1-x)}{x^2}y = 0$$

$$\Rightarrow y'' + \frac{\left(\frac{3}{2}\right)(1+2x+x^2)}{x}y' + \frac{\left(-\frac{1}{2}\right)(1-x)}{x^2}y = 0$$

[Form: $y'' + \frac{p(x)}{x}y' + \frac{q(x)}{x^2}y = 0$]

where

$p(x) = \left(\frac{3}{2}\right)(1+2x+x^2) \Rightarrow p_0 = p(0) = \frac{3}{2}$

$q(x) = \left(-\frac{1}{2}\right)(1-x) \Rightarrow q_0 = q(0) = -\frac{1}{2}$

STEP-II Indicial equation: $r(r-1) + r p_0 + q_0 = 0$

$$r(r-1) + \left(\frac{3}{2}\right)r - \frac{1}{2} = 0 \qquad [r_1 = \tfrac{1}{2}, r_2 = -1]$$

$$r^2 - r + \frac{3}{2}r - \frac{1}{2} = 0 \Rightarrow r^2 + \frac{1}{2}r - \frac{1}{2} = 0$$

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STEP - III Write the series solutions for y_1 & y_2.

Note! $\left[r_1 = \frac{1}{2}\ ,\ r_2 = -1\right]$

r_1, r_2 ⟶ Distinct roots $\quad r_1 \geq r_2$

$$\therefore \left[y_1(x) = x^{r_1} \sum_{n=0}^{\infty} a_n x^n\right] \quad (a_0 \neq 0)$$

⟶ $(r_1 - r_2) \neq 0$ & $r_1 - r_2 = \frac{1}{2} - (-1) = \frac{3}{2}$

⇒ Not a positive integer!

$$\therefore \left[y_2(x) = x^{r_2} \sum_{n=0}^{\infty} b_n x^n\right] \quad (b_0 \neq 0)$$

$$y_1(x) = x^{1/2} \sum_{n=0}^{\infty} a_n x^n$$

$$y_2(x) = x^{-1} \sum_{n=0}^{\infty} b_n x^n$$

⟶ [Two linearly independent Frobenius solutions]

EXAMPLE BESSEL'S EQUATION of ORDER ZERO

Find the Frobenius series solution of the Bessel's equation of order zero

$$x^2y'' + xy' + x^2y = 0$$

Solution:

STEP-0 Bessel's equation

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0$$

where $\nu = 0$ or a positive real number.

Here: $\nu = 0$

$\therefore x^2y'' + x\cdot y' + x^2y = 0$

Bessel's equation of order Zero

STEP-I Write the ODE in standard form:

Divide the ODE by x^2: $\frac{x^2y''}{x^2} + \frac{x}{x^2}y' + \frac{x^2}{x^2}y = 0$

$\Rightarrow \boxed{y'' + \frac{1}{x}y' + y = 0}$ form: $y'' + \frac{p(x)}{x}y' + \frac{q(x)}{x^2}y = 0$

Here:

① $x=0$ is a regular singular point

② $p(x) = 1$ $\because p(x) = x\,P(x) = x\left(\frac{1}{x}\right) = 1$

③ $q(x) = x^2$ $\because q(x) = x^2 Q(x) = x^2(1) = x^2$

④ $p(x)$ & $q(x)$ are analytic around $x=0$

STEP-2 INDICIAL EQUATION : $r(r-1)+rp_0+q_0=0$

$p(x)=1 \Rightarrow p_0=p(0)=1$

$q(x)=x^2 \Rightarrow q_0=q(0)=0$

$\Rightarrow r(r-1)+r(1)+0$

$\Rightarrow r^2-r+r=0 \Rightarrow r^2=0$

$\Rightarrow \boxed{r=0}$ → Only one exponent!

→ Only one Frobenius series solution! (NOT!)

{This is a case where $r_1=r_2$ that guarantees one Frobenius series solution

STEP-3 Write the Frobenius series solution:

$$\boxed{y(x)=x^r\sum_{n=0}^{\infty}c_n x^n}$$

↳ We derive one of the two required linearly independent solutions

$$\Rightarrow y(x)=x^0\sum_{n=0}^{\infty}c_n x^n$$

$$\therefore \boxed{y(x)=\sum_{n=0}^{\infty}c_n x^n}$$ → But this looks like a power series! (coincidentally)

STEP-4 Substitute in the ODE $[x^2y''+xy'+x^2y=0]$ and find the recurrence relation:

x^2y'' : $x^2\sum_{n=0}^{\infty}n(n-1)c_n x^{n-2}$

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$$x y' = x \sum_{n=0}^{\infty} n c_n x^{n-1}$$

$$x^2 y = x^2 \sum_{n=0}^{\infty} c_n x^n$$

$$\therefore x^2 y'' + x y' + x^2 y = \sum_{n=0}^{\infty} n(n-1) c_n x^n + \sum_{n=0}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^{n+2}$$

$$= 0$$

$$\Rightarrow \sum_{n=0}^{\infty} [n(n-1) + n] c_n x^n + \sum_{n=0}^{\infty} c_n x^{n+2} = 0$$

Combined first two terms

Need to shift index $n \to n-2$

$$\Rightarrow \sum_{n=0}^{\infty} n^2 c_n x^n + \sum_{n=2}^{\infty} c_{n-2} x^n = 0$$

$$\Rightarrow (0) c_0 x^0 + (1) c_1 x + \sum_{n=2}^{\infty} [n^2 c_n + c_{n-2}] x^n = 0$$

$\Rightarrow$ c_0 = arbitrary constant

$c_1 = 0$

$$\boxed{c_n = -\frac{c_{n-2}}{n^2}} \quad n \geq 2 \quad \{\text{Recurrence relation}\}$$

STEP-5 Find the coefficients using

$$\left[C_n = -\frac{C_{n-2}}{n^2}\right] \quad \& \; C_1 = 0$$

$$C_2 = \frac{-C_0}{2^2} \qquad C_3 = -\frac{C_1}{3^2} = 0$$

$$C_4 = \frac{-C_2}{4^2} = \frac{C_0}{2^2 4^2} \qquad C_5 = -\frac{C_3}{5^2} = \frac{C_1}{3^2 5^2} = 0$$

$$C_6 = \frac{-C_4}{6^2} = \frac{-C_0}{2^2 4^2 6^2} \qquad C_7 = \frac{-C_5}{7^2} = \frac{-C_1}{3^2 5^2 7^2} = 0$$

$$\vdots \qquad\qquad \vdots$$

$$C_{2n} = \frac{(-1)^n C_0}{2^2 4^2 \cdots (2n)^2} \qquad C_n = 0 \text{ for all odd } n$$

$$\Rightarrow \boxed{C_{2n} = \frac{(-1)^n C_0}{2^{2n} (n!)^2}}$$

STEP-6 Final answer for one of the two Frobenius series solutions of the Bessel's equation of order zero.

$$\{J_0(x)\}$$

$$y_1(x) = \sum_{n=0}^{\infty} C_{2n} x^{2n} = J_0(x)$$

$$\Rightarrow \boxed{J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n} (n!)^2} x^{2n}} \quad \{\text{where } C_0 = 1\}$$

$$= 1 - \frac{x^2}{4} + \frac{x^4}{64} - \frac{x^6}{2304} + \cdots$$